

PySpark ETL Project Assignment

Retail Black Friday Sales Analytics Platform

Assignment Duration

2 Hours

Platform Requirement

Mandatory Platform

This assignment must be completed using:

- **Google Colab only**

Local IDEs, Jupyter Notebook, or other environments are not allowed.

Objective

You are provided with the dataset:

Dataset File

`retail_black_friday_sales_100k.csv`

Your task is to design and implement a **PySpark-based ETL Pipeline** to process, clean, transform, analyze, and visualize Black Friday retail sales data.

This assignment simulates a real-world Retail Data Engineering workflow where raw customer transaction data is transformed into business-ready analytical insights.

Technical Requirements

Mandatory Technologies

- Python
- PySpark
- Spark SQL
- PySpark DataFrame API

Visualization Libraries Allowed

(Only after transformations are completed)

- Matplotlib
- Plotly
- Seaborn

Restrictions

Do NOT Use

- Pandas for ETL transformations
- SQL databases
- External ETL tools/platforms

Pandas may only be used optionally for converting final aggregated outputs for visualization purposes.

ETL Workflow

Phase 1 — Extract

Load the dataset into PySpark.

Tasks

1. Initialize Spark Session in Google Colab
 2. Define schema explicitly
 3. Load CSV properly
 4. Handle malformed records
 5. Validate null values
 6. Print schema and sample records
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Phase 2 — Transform

Perform all required transformations.

Data Cleaning Tasks

- Remove duplicate records
 - Handle missing/null values
 - Normalize sales-related columns
 - Standardize categorical text columns
 - Convert transaction dates into proper date datatype
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Feature Engineering

Create the following derived columns:

Column Name	Description
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Discount_Percentage	Percentage discount applied
Revenue_Per_Customer	Revenue generated per customer
High_Value_Order_Flag	1 if order exceeds threshold
Weekend_Sale_Flag	1 if transaction occurred on weekend
Product_Category_Count	Number of product categories purchased

Advanced Transformations

- Aggregate category-level sales
 - Build customer purchase frequency tables
 - Create city-level sales statistics
 - Rank top-selling products using window functions
 - Identify peak sales periods
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Phase 3 — Load

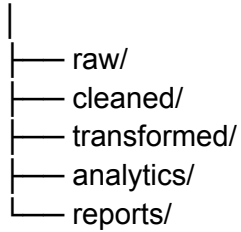
Store transformed outputs into structured folders.

Output Formats

- Parquet
 - CSV summaries
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Suggested Output Structure

output/



Use Case 1 — Sales Performance Analytics

Business Problem

Retail companies want to identify:

- Top-performing product categories
 - Cities generating maximum revenue
 - Peak shopping periods during Black Friday sales
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Required ETL Tasks

Perform the following:

- Category-wise sales aggregation
 - City-wise revenue analysis
 - Revenue trend analysis
 - Product ranking by sales volume
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Spark Concepts Expected

- `groupBy()`
- `agg()`
- `orderBy()`

- Window Functions
 - `dense_rank()`
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Visualizations Required

Visual 1 — Top Product Categories by Revenue

Suggested Visualization:

- Bar Chart
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Visual 2 — Top Cities by Revenue

Suggested Visualization:

- Horizontal Bar Chart
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Visual 3 — Sales Trend Heatmap

Suggested Visualization:

- Heatmap of:
 - Time vs Revenue
 - Category vs Sales
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Use Case 2 — Customer Purchase Intelligence

Business Problem

Marketing teams want to analyze:

- High-value customers

- Customer purchasing behavior
 - Discount effectiveness
 - Repeat purchase trends
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Required ETL Tasks

Perform the following:

- Analyze customer purchase frequency
 - Compute average spend per customer
 - Compare discounted vs non-discounted sales
 - Identify repeat customers
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Spark Concepts Expected

- `withColumn()`
 - Aggregations
 - Spark SQL
 - Conditional Transformations
 - Window Functions
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Visualizations Required

Visual 1 — Top Customers by Revenue

Suggested Visualization:

- Bar Chart
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Visual 2 — Discount vs Sales Distribution

Suggested Visualization:

- Histogram

Visual 3 — Customer Purchase Frequency Analysis

Suggested Visualization:

- Pie Chart / Donut Chart
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Use Case 3 — Product & Inventory Insights

Business Problem

Retail operations teams want to understand:

- Fast-moving products
 - Inventory-demand patterns
 - Seasonal buying behavior
 - Product category growth
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Required ETL Tasks

Perform the following:

- Detect high-demand products
 - Analyze category growth trends
 - Compute rolling sales averages
 - Identify seasonal purchasing patterns
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Spark Concepts Expected

- `date_format()`
- Window Functions
- Rolling Aggregations

- Time-series transformations
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Visualizations Required

Visual 1 — Monthly Sales Trends

Suggested Visualization:

- Line Chart
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Visual 2 — Fastest Growing Categories

Suggested Visualization:

- Multi-Line Trend Graph
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Visual 3 — Seasonal Purchase Distribution

Suggested Visualization:

- Area Chart
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Bonus Requirements (Optional)

Implement any 5 of the following:

Performance Engineering

- Repartitioning
- Caching
- Broadcast joins
- Predicate pushdown
- Partition pruning

Advanced Spark Features

- Spark SQL Views
 - UDF creation
 - Structured logging
 - Incremental ETL
 - Spark MLlib integration
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Data Quality Engineering

- Validation checks
 - Schema enforcement
 - Outlier detection
 - Duplicate tracking
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Deliverables

Code Deliverables

Submit:

- Google Colab Notebook ([.ipynb](#))
 - ETL scripts
 - Visualization notebook
 - Spark SQL queries
 - Final transformed outputs
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Documentation Deliverables

Include:

Architecture Explanation

- Data ingestion flow
 - Transformation stages
 - Analytical outputs
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ETL Documentation

- Cleaning decisions
 - Feature engineering logic
 - Spark optimization methods
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Business Insights Report

Summarize:

- Sales trends
 - Customer intelligence
 - Product demand
 - Revenue insights
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Evaluation Criteria

Criteria	Weightage
PySpark ETL Design	25%
Transformation Logic	20%
Spark Optimization	15%

Visualizations	15%
Business Insights	15%
Code Quality	10%

Expected Learning Outcomes

After completing this assignment, learners should understand:

- Real-world ETL architecture
- Distributed data processing using Spark
- Spark SQL transformations
- Retail analytics engineering workflows
- Customer intelligence systems
- Data visualization pipelines
- Performance optimization in Spark

[Dataset Link](#)